ALIEN-DL

Adaptive Local Interaction and Evolving Networks

A resource-coupled synaptic architecture for deep learning
Two independent implementations, code-level audit,
and initial small-data experimental evidence



RESEARCH TRIALS REPORT / V1.0

Devanik

Project lead · AI-assisted research and drafting support disclosed

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GitHub source: The-Invention-Archive

01 · EXECUTIVE SUMMARY

Abstract and evidence ledger

The report reconstructs the two committed implementations and evaluates what their first small-data trials do - and do not - establish.

Abstract. ALIEN-DL is an exploratory family of artificial neural network prototypes inspired by biological synapses as dynamic, stateful computational pathways. The shared hypothesis is that connection efficacy should depend not only on a trainable scalar but also on short-term resource availability, facilitation, local allocation and, in one implementation, structural persistence. Two independently coded trials were audited: Trial A, a PyTorch model with eight dendritic-style compartments; and Trial B, a compact NumPy/autograd model with one hidden layer and synapse-specific transient dynamics. Both were compared with conventional multilayer perceptrons on small benchmark datasets. Trial A reached 98.45% mean accuracy on Breast Cancer, matching its parameter-matched MLP, but trailed it on Wine and Digits; it trained approximately 3.6x-6.3x more slowly. Trial B showed allocation instability and lower Digits accuracy under its default adaptive settings. Neither trial demonstrates a superadditive benefit from the proposed coupling, nor a consistent predictive advantage. Both implementations reset short-term synaptic states on each independent forward call, so the experiments do not test persistent temporal memory. The present contribution is therefore a transparent prototype family and a falsifiable research programme, not a validated new learning paradigm.

Evidence item	Trial A · PyTorch	Trial B · lite
Architecture	8 dendritic-style compartments; structural gate	1 hidden layer (H=16); no topology state
Adaptive allocation	Heuristic contribution score; epoch update	Autograd gradient w.r.t. allocation; batch update
R / P / A semantics	Recovery / plasticity / structural persistence	Recovery / plasticity / adaptive allocation
Main comparison	Wine, Breast Cancer, Digits; 3 seeds	Breast Cancer, Digits; 3 seeds; 40 epochs
Outcome	Tie on Breast Cancer; below matched MLP elsewhere	Below MLP baselines on tested Digits settings
Temporal memory	Not tested; r,u reset per forward batch	Not tested; r,u reinitialized per forward call

Bottom line

The first tests establish that synapse-specific dynamic states and non-uniform allocation can be implemented and trained. They do not establish that the combined ALIEN mechanisms improve generalization, sample efficiency, temporal memory, robustness or efficiency.

Evidence labels used throughout. “Measured” refers to values in the committed Trial A CSV or the Trial B README summary. “Code-audited” refers to behavior visible in the implementation. “Proposed” refers to the next mathematical design, not an evaluated feature. Trial B per-seed JSON/log artifacts are described as residing in the ZIP bundle; the extracted GitHub folder exposes its source code but not the per-seed result files separately.

02 · SCIENTIFIC MOTIVATION

Why change the ANN abstraction?

The motivating question is architectural: what changes when a connection is treated as a stateful dynamical object rather than a static scalar parameter?

The standard feed-forward unit computes an affine aggregation followed by a nonlinearity:

Conventional point-neuron abstraction

$$h_j = \sigma(\Sigma_i w_{ij} x_i + b_j)$$

This abstraction is useful and mathematically tractable. It deliberately compresses many biological processes into a weight and an activation. In biological circuits, however, synapses can differ in strength, location, depression/recovery, facilitation and plasticity state; dendritic compartments can perform local nonlinear integration; and neighboring synapses can compete for finite molecular building blocks [2-5]. These are not arguments that biological detail automatically improves machine learning. They motivate testable alternative parameterizations.

Biological observation	Computational analogue	Caution
Short-term depression and facilitation	Per-edge r and u state variables	The code uses smooth micro-step approximations, not full spike-event models.
Local resource competition	Normalized allocation share pi	The implemented pi is an abstract normalized multiplier, not a measured biochemical pool.
Dendritic compartments	Local integration domains	Only Trial A includes explicit multi-compartment aggregation.
Structural plasticity	Edge persistence / pruning gate	Only Trial A has an edge mask and thresholded death; neither has birth.
Heterogeneous plasticity	Allocation-gated parameter updates	Both trials still use global loss gradients for task learning.

Design principle

ALIEN-DL does not attempt to simulate a complete biological neuron. It asks whether selected synapse-level state variables, when coupled to transmission and optimization, create useful computational behavior.

The biological analogy has boundaries. The Trial A and B state variables are computational proxies. They should not be described as literal neurotransmitter vesicle counts, receptor concentrations, calcium concentrations or astrocytic variables. The current models are deterministic, low-dimensional dynamical networks trained on static tabular samples.

03 · PROPOSED ARCHITECTURE

ALIEN-DL as a resource-coupled synaptic system

The intended architecture couples three pathways to a local allocation state. The two prototypes instantiate different subsets of this idea.

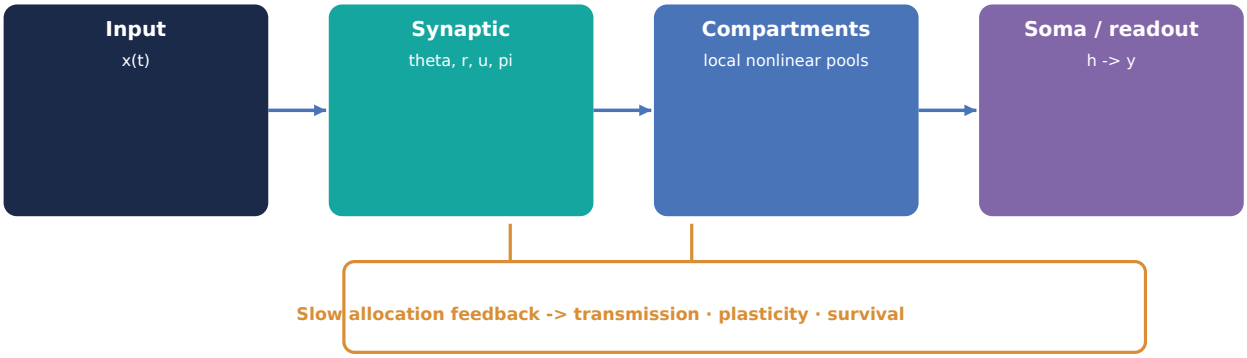


Figure 1. Conceptual design target. A synapse carries trainable efficacy and transient state; local compartments integrate incoming signals; a slow allocation process can influence three distinct pathways. The drawing is a conceptual synthesis, not a literal diagram of either implementation.

Pathway	Intended role	Trial A	Trial B
R · transmission	Allocation changes the recovery dynamics of r.	Implemented	Implemented
P · plasticity	Allocation gates the synaptic efficacy update.	Implemented	Implemented
A · structure	Allocation affects persistence / edge survival.	Implemented	Not implemented as structure
Adaptive allocation	The allocation distribution is learned.	Always learned using a heuristic score	Enabled by A switch using autograd gradient

The three switch letters therefore do not mean precisely the same thing in both trials. In Trial A, A toggles the structural gate and pruning update; the allocation logits are still updated by the contribution heuristic. In Trial B, the code variable `Ac` gates adaptive allocation itself, and there is no structural persistence variable or pruning routine. This semantic difference is essential when comparing the factorial results.

Interpretive rule

The two trials are related explorations of a common hypothesis, not identical implementations of one fixed mathematical model. Their R/P/A labels must be read according to each codebase.

04 · TRIAL A

PyTorch prototype: compartmentalized synaptic dynamics

Source: `A_xylia_prototype/alien_dl.py`. The model is a classifier with eight dendritic-style compartments and a linear class readout.

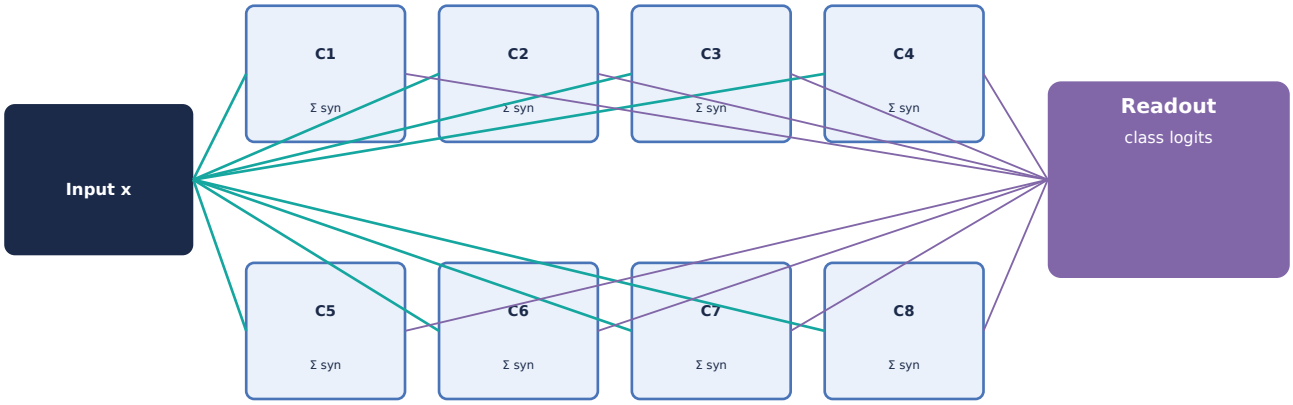


Figure 2. Trial A signal path. Every compartment receives a feature-wise set of synapses. Synaptic outputs are summed locally, passed through a compartment nonlinearity and combined by a readout. Short-term states are created for the current batch; allocation logits persist as trainable model state.

Per-edge state. For edge e from feature i into compartment k, the code stores trainable efficacy θ_e, allocation logit ℓ_e, transient available-resource state r_e, transient facilitation state u_e, and a structural logit ζ_e with a binary mask m_e. The configuration defaults to eight compartments and three micro-steps.

Mean-one allocation over active synapses

$$\pi_{e,k} = n_k \cdot \mathrm{softmax}(\ell_{\cdot,k})_e,\ \Sigma_{e \in S_k}\, \pi_{e,k} = n_k$$

Effective transmission

$$q_e = \theta_e \cdot u_e \cdot r_e \cdot a_e \cdot \tanh(x_i)$$

Structural multiplier

$$a_e = m_e \cdot \mathrm{sigmoid}(\zeta_e)\ \text{(A on)};\ a_e = m_e\ \text{(A off)}$$

Here n_k is the number of active synapses in the compartment and S_k is the active set. The allocation is normalized to the current active fan-in; it is not an independently conserved physical resource pool.

05 · TRIAL A

Dynamics, learning rules and topology events

The PyTorch code unrolls three internal updates using the same static feature vector, then forms compartment outputs.

R pathway · allocation-coupled recovery

$$r \leftarrow \mathrm{clip}[\, r + \rho \cdot \pi_e (1 - r) - \alpha_r \cdot u \cdot r \cdot |\tanh(x_i)|,\ 0,\ 1\ ]$$

Facilitation-like state update

$$u \leftarrow \mathrm{clip}[\, u + (u_0 - u)/\tau_u + \alpha_u (1 - u) |\tanh(x_i)|,\ 0,\ 1\ ]$$

Compartment integration and classifier readout

$$d_k = \tanh\Big(\sum_{e \in S_k} q_e + b_k \Big);\quad y = W_{\mathrm{readout}} d + b_{\mathrm{out}}$$

Allocation utility. During each minibatch, the code differentiates loss with respect to the transmitted edge signal q and forms a contribution heuristic:

Heuristic contribution score

$$G_e = -\, q_e \cdot \partial L / \partial q_e$$

At epoch end the row-wise score is normalized by mean absolute magnitude, centered relative to the current allocation distribution and added to the allocation logits. This is a logit-space, multiplicative-weights-inspired heuristic. It is not the exact gradient ∂L/∂b of a conserved physical resource budget.

P pathway · allocation-gated efficacy update

$$\theta_e \leftarrow \theta_e - \eta_\theta \cdot \pi_e \cdot \partial L / \partial \theta_e \ \text{(P on)}$$

A pathway · persistence integrator and hard threshold

$$\zeta_e \leftarrow \mathrm{clip}[\, \zeta_e + \kappa(\pi_e - 1),\ -8,\ 8\ ];\ \text{prune if } \mathrm{sigmoid}(\zeta_e) < \tau_p$$

The structural step runs at epoch boundaries and preserves at least one incoming edge per compartment. Structural births are absent. The transient r and u arrays are initialized inside each forward pass; they do not persist from one independent training example to the next. The code uses ordinary backpropagation for the task gradients, with manually gated θ updates; it is not a fully local biological learning rule.

06 · TRIAL A RESULTS

Small-data benchmark: three datasets, three seeds

Reported protocol: stratified 70/15/15 train/validation/test splits; training-only standardization; seeds 7, 21 and 42; validation-best checkpoint evaluated once on the held-out test set.

Dataset scale. Wine: 178 samples, 13 features, 3 classes; Breast Cancer Wisconsin: 569 samples, 30 features, 2 classes; Digits: 1,797 samples, 64 pixel features, 10 classes. These are small benchmark datasets, not large-scale deep-learning workloads.

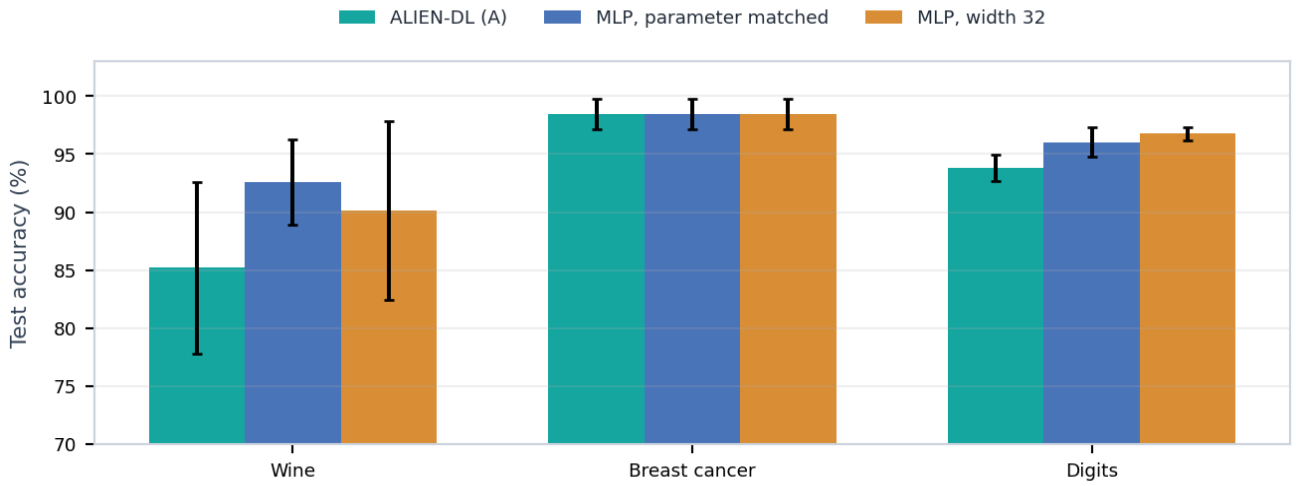


Figure 3. Mean test accuracy ± standard deviation across three seeds. The parameter-matched MLP uses a nearby trainable-parameter count; the width-32 MLP is a wider reference. Values are from Trial A per-seed benchmark CSV.

Dataset	Model	Accuracy	Macro-F1	Params	Train time
Wine	ALIEN-DL	85.19 ± 7.41%	84.14 ± 8.28%	243	0.164 s
Wine	MLP matched	92.59 ± 3.70%	92.66 ± 3.70%	241	0.046 s
Wine	MLP width 32	90.12 ± 7.71%	90.25 ± 7.61%	547	0.042 s
Breast Cancer	ALIEN-DL	98.45 ± 1.34%	98.33 ± 1.45%	506	0.599 s
Breast Cancer	MLP matched	98.45 ± 1.34%	98.36 ± 1.42%	497	0.157 s
Breast Cancer	MLP width 32	98.45 ± 1.34%	98.36 ± 1.42%	1,058	0.120 s
Digits	ALIEN-DL	93.83 ± 1.13%	93.85 ± 1.17%	1,122	2.678 s
Digits	MLP matched	96.05 ± 1.30%	96.04 ± 1.30%	1,135	0.424 s
Digits	MLP width 32	96.79 ± 0.57%	96.78 ± 0.57%	2,410	0.461 s

Interpretation

Trial A ties its parameter-matched MLP on Breast Cancer, while trailing it by 7.41 percentage points on Wine and 2.22 points on Digits. The three-seed small-sample results are descriptive; the Wine standard deviations are large relative to the mean differences.

07 · TRIAL A MECHANISMS

Allocation concentration and 2 × 2 × 2 ablation

The main checkpoints retained all candidate synapses. The allocation distribution nevertheless became non-uniform.

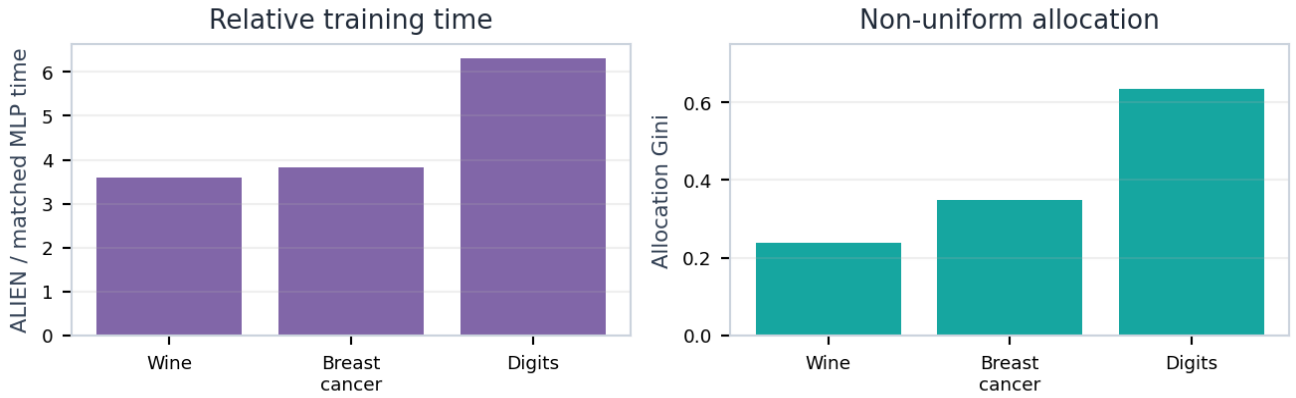


Figure 4. Left: CPU training time divided by parameter-matched MLP training time. Right: mean allocation Gini across three seeds. Gini indicates concentration/inequality, not useful specialization by itself.

Mean allocation Gini. Wine = 0.2375 (SD 0.0476); Breast Cancer = 0.3483 (SD 0.1046); Digits = 0.6352 (SD 0.1192). The effective fan-in statistic also decreases as allocation concentrates, but it should be interpreted cautiously where pruning changes the active set.

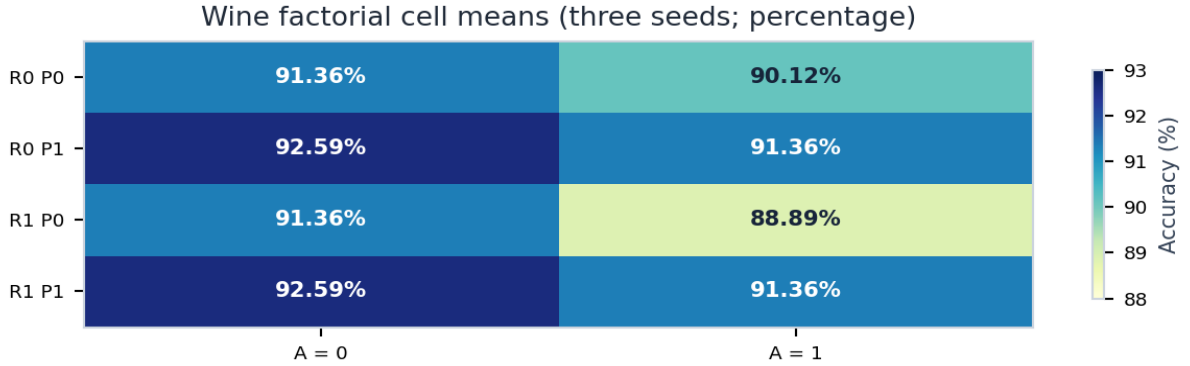


Figure 5. Trial A Wine factorial cell means. Each cell is averaged over three seed runs; the Wine held-out test partition contains 27 cases per seed.

R	P	A	Mean accuracy	SD
0	0	0	91.36%	8.55%
0	0	1	90.12%	10.69%
0	1	0	92.59%	9.80%
0	1	1	91.36%	11.91%
1	0	0	91.36%	8.55%
1	0	1	88.89%	9.80%
1	1	0	92.59%	9.80%
1	1	1	91.36%	11.91%

Conditional interactions recomputed from the per-seed CSV. At fixed levels: R×P | A=0 = 0; R×A | P=0 = −0.01235 (SD 0.02138); P×A | R=0 = 0. The three-way contrast is +0.01235 (SD 0.02138). At the alternate fixed levels, R×P | A=1 and P×A | R=1 are +0.01235; R×A | P=1 is 0. Every non-zero seed contrast equals one Wine test case, 1/27 = 0.03704.

Statistical reading

Three seeds and 27 test examples per split cannot establish superadditivity. These conditional contrasts are descriptive and highly quantized, not inferential evidence of a coupling advantage.

08 · TRIAL B

Lite NumPy/autograd prototype: a synapse-state hidden layer

Source: `B_lite_trial/code/alien.py`, `run_alien.py`, `eta_sweep.py` and `unit_tests.py`. Trial B is deliberately more compact than Trial A.

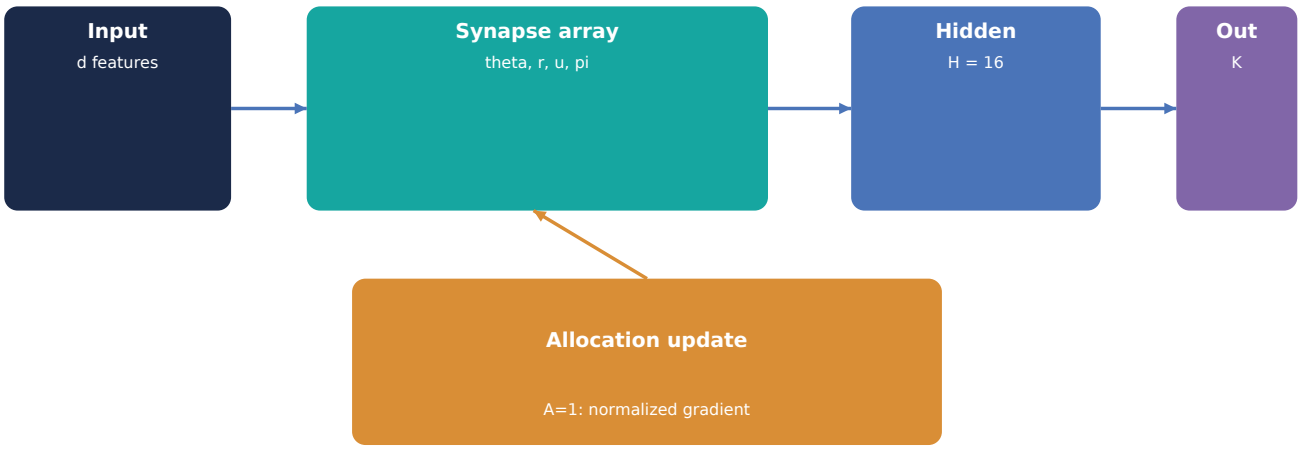


Figure 6. Trial B has an input-to-hidden synapse array, a standard tanh hidden layer and a linear output. It does not contain Trial A's eight dendritic-style compartments or a structural mask.

Column-wise mean-one allocation per hidden unit

$$\pi_{ij} = d \cdot \mathrm{softmax}(\lambda_{\cdot j})_i,\ \Sigma_{i=1}^{d}\, \pi_{ij} = d$$

R pathway · recovery dynamics

$$r_{ij} \leftarrow \mathrm{clip}[\, r + \Delta t(\rho \cdot \pi_{ij}^{R}(1-r) - \alpha_r u\, r\, x_i),\ 0,\ 1\,]$$

Facilitation-like dynamics

$$u_{ij} \leftarrow \mathrm{clip}[\, u + \Delta t((u_0-u)/\tau_u + \alpha_u(1-u)x_i),\ 0,\ 1\,]$$

A pathway · normalized allocation-logit update

$$\lambda \leftarrow \lambda - \eta_b \cdot \nabla_\pi L \,/\, (\mathrm{mean}|\nabla_\pi L| + \varepsilon)$$

Hidden-unit integration (T = 5 micro-steps)

$$h_j = \tanh[\, (1/d)\ \Sigma_i\, \theta_{ij}\, x_i \cdot \Sigma_{t=1}^{T}\, r_{ij,t} u_{ij,t} + b_j\,]$$

The code normalizes inputs to [0,1] using training-set min/max values. It has 16 hidden units by default and performs five internal updates using the same static sample. `r` and `u` are reinitialized on each forward call. The model therefore introduces within-pass dynamics, not persistent sample-to-sample memory. The P pathway multiplies the θ gradient before Adam; because Adam maintains adaptive first- and second-moment estimates, this is not identical to a direct per-edge learning-rate multiplier and should be analyzed as gradient modulation. The H=32 repository-labelled state-matched MLP has roughly twice the trainable parameters of the H=16 ALIEN model on these datasets; the H=16 plain MLP is the closer parameter-count comparator.

09 · TRIAL B EVIDENCE

Adaptive allocation, baseline results and collapse

The README summarizes three seeds for Breast Cancer and Digits, with a 70/30 train/test split, 40 epochs and H=16. Trial B does not have a validation partition in the committed runner.

Operational meaning of switches. R=1 allows allocation shares to modulate resource recovery; R=0 uses a unit recovery multiplier. P=1 multiplies the θ gradient by π before Adam; P=0 leaves it unscaled. A=1 enables the adaptive allocation-logit updates; A=0 fixes π to uniform. There is no structural survival variable, hard pruning, edge birth or graph topology update in this code.

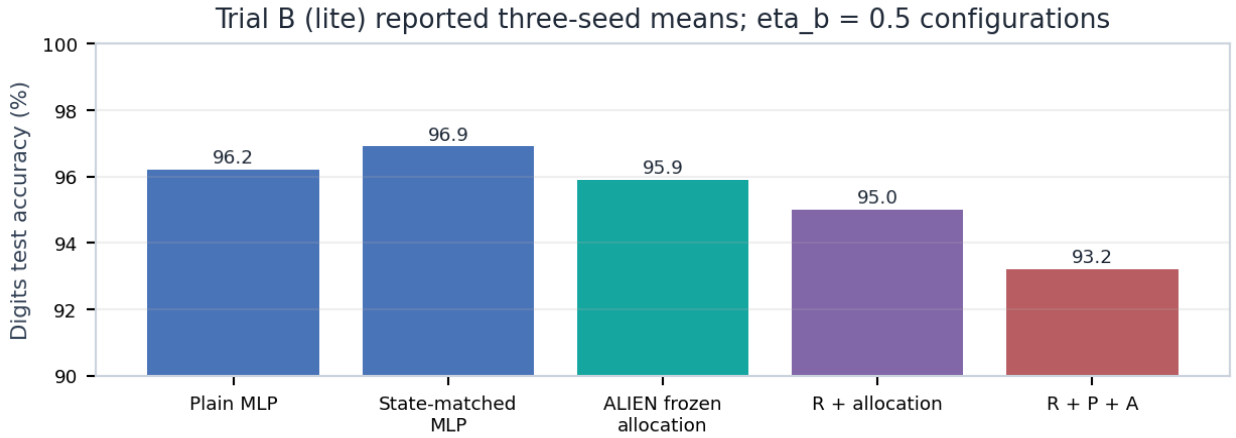


Figure 7. Trial B Digits means transcribed from the repository README summary. The reported default configuration uses η_b=0.5; no per-configuration standard deviations are provided in the extracted README.

Dataset / configuration	Reported mean accuracy	Interpretation
Digits · plain MLP (H=16)	0.962	Reference baseline
Digits · repository state-matched MLP (H=32)	0.969	Wider state-budget reference
Digits · ALIEN, frozen allocation	0.959	Near plain MLP; below state-matched MLP
Digits · R + adaptive allocation	0.950	Allocation concentration can harm fit
Digits · R + P + A	0.932	Lowest listed configuration
Breast Cancer · frozen / most coupled	0.961	Repository summary; exact per-cell values not expanded
Breast Cancer · full coupling	0.949	Below the other listed settings

Allocation collapse

At η_b = 0.5, the repository summary reports a single synapse receiving approximately 63 of the 64 allocation units in a hidden-unit column. The η sweep used the test split for selection; the lower-step diagnostic is therefore exploratory and must not be presented as a validation-independent result.

The repository README reports that at η_b=0.05 the largest share stayed near 6.8 and that the R + allocation Digits accuracy recovered to about 0.960. Since the sweep selected on the test data, this is a tuning diagnostic only. The bundled wine.json is explicitly marked as an earlier timing run before rate and step-size fixes and is excluded from cross-trial quantitative conclusions.

10 · CROSS-TRIAL SYNTHESIS

Two implementations, different semantics and protocols

The trial outputs should not be pooled into a single ALIEN accuracy estimate. Architecture, preprocessing, training, validation and coupling semantics differ.

Dimension	Trial A · PyTorch	Trial B · NumPy/autograd
Core architecture	8 local compartments + readout	Single hidden layer H=16 + readout
Synaptic transient state	r and u per batch × compartment × feature	r and u per call × feature × hidden unit
Internal steps	3 micro-steps	5 micro-steps
Allocation shape	Per compartment; sum equals active fan-in	Per hidden unit; sum equals input dimension d
Allocation signal	Normalized −q·dL/dq heuristic; epoch update	Autograd ∂L/∂π; normalized; minibatch update
θ learning	Manual SGD with allocation multiplier when P=1	Adam update with θ gradient multiplied by π when P=1
Structure	ζ persistence gate + edge mask; pruning only	No structural state or topology operations
R/P/A meaning	Recovery / θ update / structural persistence	Recovery / θ update / adaptive allocation
Data protocol	70/15/15; training-only standardization; validation checkpoint	70/30; train-fitted min-max; no validation split
Datasets	Wine, Breast Cancer, Digits	Breast Cancer, Digits; Wine JSON non-comparable
Seed count	3 (7, 21, 42)	3 (0, 1, 2)

Why no pooled score?

A pooled score would conflate different feature scaling, split ratios, optimization rules, output state, and allocation semantics. The only defensible cross-trial synthesis is qualitative: both implement non-uniform synaptic allocation, both expose allocation instability, and neither currently establishes a reliable advantage over its conventional MLP baseline.

A meaningful next comparison would require a single shared data pipeline, identical splits, identical evaluation metrics, fixed model-selection rules, matched parameter and state budgets, and a common definition of the allocation and structural mechanisms.

11 · MATHEMATICAL AUDIT

What is conserved in the current code?

The v2 theory proposed a fixed budget Bk independent of the number of edges. Neither prototype implements that exact physical-ledger formulation.

Trial A. For each compartment k, the code applies softmax to active edge logits and multiplies by the active fan-in n_k:

Implemented normalization

$$\Sigma_{e \in S_k} \pi_{e,k} = n_k$$

If an edge is removed, n_k changes and the sum of the dimensionless allocation multipliers changes accordingly. The code does redistribute the normalized share over survivors, but the invariant is “mean allocation equals one over active edges,” not “a fixed amount of physical resource is conserved independently of topology.”

Trial B. For hidden unit j, the same mean-one convention applies across d inputs:

Implemented normalization

$$\Sigma_{i=1}^{d} \pi_{ij} = d$$

Since B has fixed input dimension, the sum is constant during a run. It is still a dimensionless normalization of shares, not an explicit conserved resource pool that changes consistently with structural births and deaths.

Proposed fixed-ledger design for a future version - not implemented

$$b_e \geq 0,\ \Sigma_{e \in S_k} b_e = B_k,\ \pi_e = n_k b_e / B_k$$

Optimization identity. The continuous replicator equation `dp_i/dt = η p_i(U_i − Ū)` is a known population-dynamics / multiplicative-weights form. It is not itself a novel update rule. In Trial A, logits are updated by a centered heuristic score; in Trial B, logits are updated by a normalized autograd gradient with respect to π. These code paths are distinct from a direct exponentiated-gradient step in b.

Important claim boundary

The present experiments test normalized allocation-dependent dynamics. They do not test the v2 “single conserved physical resource ledger” as mathematically specified. The fixed-ledger equations belong to the next design iteration.

12 · EVIDENCE AND KILL-CRITERIA AUDIT

What the experiments support - and what they do not

A research report should make negative and inconclusive evidence visible, not bury it under an architecture narrative.

Statement	Status	Reason
Synaptic allocation can become non-uniform	Supported in both trials	Trial A Gini > 0; Trial B share concentration changes with ηb.
Dynamic synaptic state can be inserted into a classifier	Supported as an implementation fact	Both forward passes use r and u state variables.
ALIEN consistently beats parameter-matched MLPs	Not supported	Trial A ties only on Breast Cancer; below on Wine and Digits. Trial B below listed MLPs on Digits.
Three mechanisms yield superadditive benefit	Not supported	Trial A three-way mean contrast is +0.01235 with SD 0.02138; one test-case step.
Structural plasticity improves performance	Not established	Trial A pruning is sparse/inconsistent; Trial B has no structural mechanism.
Temporal memory improves	Not tested	r and u are reset on each independent forward call.
Computational efficiency improves	Contradicted in Trial A run	3.6x-6.3x slower than matched MLP on CPU.
Biological fidelity has been demonstrated	Not supported	State variables are abstract deterministic proxies.

Pre-specified kill criterion from the v2 plan. If interaction contrasts fail to show a robust non-zero effect, retract the superadditivity claim. The present Trial A data do not meet a defensible evidential threshold: three seeds, small Wine test sets, discrete accuracy steps and no corrected inferential analysis. The correct conclusion is “no demonstrated superadditive interaction,” not “proof of no interaction.”

Architecture claim. The prototypes are an identifiable design family, but the individual ingredients have prior art: dendritic ANN computation, synaptic diversity, short-term plasticity, synaptic resource competition, state-dependent plasticity and exponentiated-gradient synaptic learning. The paper therefore makes no priority claim that the synapse-centric idea or any ingredient is new.

Current result

ALIEN-DL is a working exploratory prototype family with measurable allocation dynamics and a clear set of failures to address. It is not yet a validated alternative to conventional deep learning.

13 · THREATS TO VALIDITY

Limitations and implementation-level caveats

The following limitations materially constrain interpretation of the first experiments.

Threat	Consequence for interpretation	Required remedy
Very few seeds (n=3)	Unstable means and standard deviations; low inferential power.	At least 10-20 seeds for follow-up; paired seed design.
Small held-out Wine set (27 cases)	Accuracy changes in steps of 1/27; factorial effects are coarse.	Repeated nested CV or larger independent benchmark.
Trial B has no validation split	Model settings / η sweep risk use of test information.	Separate train/validation/test; choose settings on validation only.
η sweep uses test split	The low-step accuracy is tuned evidence, not unbiased evaluation.	Preregister η grid and select on validation; final test once.
Different A-switch semantics	Factorial results are not same experimental factor across A and B.	Define common R/P/A API and shared semantics.
Reset transient state	No persistent sequence-level synaptic memory is tested.	Stateful stream runner with reset boundaries explicitly controlled.
Static input repeated over micro-steps	Internal relaxation is not an event-driven sequence or spike train.	Feed time-varying events with defined Δt and state transition.
No births; limited pruning	Morphodynamic topology is incomplete.	Add controlled birth/death with exact budget accounting.
Allocation collapse	A simplex can concentrate almost all mass on one edge.	Entropy floor, temperature, trust region or KL constraint.
Runtime overhead	More state may cost more than it contributes.	Profile FLOPs, memory, wall-clock and throughput.
No environment lockfile	Exact dependency reconstruction is not guaranteed.	Freeze Python, PyTorch, NumPy, autograd and sklearn versions.

A further distinction: “parameter matched” is not the same as “state matched.” Trial A matches trainable parameter counts but carries additional transient state and per-edge dynamics. Trial B compares a hidden-width-16 MLP and a width-32 reference, but the width-32 MLP has more trainable parameters than the ALIEN hidden-16 model. A decisive efficiency claim requires matching trainable parameters, persistent state memory, transient activations and compute budget separately.

14 · NEXT EXPERIMENT

A temporal-memory benchmark is the correct next test

The initial motivation concerns synaptic memory and history-dependent communication. Static tabular classification does not probe that property.

Proposed benchmark. Construct a deterministic sequence classification task with two pulse symbols, A and B, where the label depends on order rather than total input. Each sequence has matched total activation and matched symbol counts; only order differs. A second variant inserts distractor pulses and variable gaps to measure state retention and interference.

Stateful temporal computation

$$s_{t+1} = F(s_t, x_t; \theta, \pi), \; y = G(s_{1:T})$$

Protocol. Compare: (i) MLP on temporally aggregated inputs, (ii) recurrent baseline (GRU or simple RNN), (iii) Trial A modified to preserve per-edge r/u states across time steps, and (iv) Trial B modified to preserve its per-edge states across sequence steps. Include state-reset and state-shuffle controls. Use identical sequences and train/validation/test splits for every model.

Primary outcomes. Accuracy by sequence length, temporal-order discrimination, performance versus distractor count, state decay / recovery curves, calibration, memory footprint and runtime. Secondary mechanism diagnostics: allocation entropy, Gini, effective fan-in, per-edge state autocorrelation, structural survival and sensitivity to state reset.

Experiment	Question answered	Control
Order inversion	Does network distinguish AB from BA?	Same token counts and total magnitude
Variable delay	How long does synaptic state retain useful information?	Delay-only control with state reset
Distractor insertion	Can relevant traces survive intervening inputs?	Matched number / energy of distractors
State ablation	Do r/u variables matter?	Reset r/u after every time step
Allocation ablation	Does learned allocation help memory?	Frozen uniform π; shuffled π
Budget ablation	Does fixed resource accounting matter?	Mean-one fan-in normalization

Decision gate before the next paper

Do not expand the model or claim temporal memory until the temporal-order task, state-reset controls and parameter/state-matched baselines are complete. A positive result must replicate across seeds and remain after validation-only hyperparameter selection.

15 · LITERATURE POSITIONING

Prior art and the bounded novelty claim

The field already contains close biological-to-ANN transfers. The present report distinguishes established mechanisms from the specific coupling hypothesis that remains to be tested.

Work	Established contribution relevant to ALIEN-DL	Implication
Markram, Wang & Tsodyks (1998) [2]	Synapses from the same axon can show target-specific depression/facilitation and recovery.	Short-term synaptic dynamics are established biology.
Triesch, Vo & Hafner (2018) [3]	Competition for local synaptic building blocks can produce multiplicative scaling and heterosynaptic effects.	Resource competition is not a new biological premise.
Chistiakova et al. (2015) [4]	Review of heterosynaptic plasticity models and homeostatic roles.	Competition and homeostasis have a mature literature.
Chavlis & Poirazi (2025) [6]	Dendritic ANN architecture with robust and parameter-efficient learning results.	Dendritic ANN design is prior art.
Hofmann et al. (2025) [7]	Transfers synaptic diversity, spine remodeling and multisynaptic connectivity into ANNs.	Synaptic diversity and structural remodeling in ANN are prior art.
Kosowski et al. (2025) [8]	BDH uses locally interacting neuron particles and Hebbian synaptic plasticity for inference-time working memory.	Synaptic plasticity as a computational memory mechanism is prior art.
Yao et al. (2026) [9]	Connects dendritic computation with foundation-model attention and proposes a dendritic-inspired hybrid attention model.	Dendritic computation has reached foundation-model architectures.
Turcu et al. (2026) [10]	psRNN routes local Hebbian-like and non-local BPTT signals according to synaptic state.	State-dependent learning-signal routing is close prior art.
Cornford et al. (2026) [11]	Exponentiated-gradient learning yields brain-like synaptic distributions, pruning robustness and performance on sparse-relevance tasks.	Gradient/EG-based brain-like synaptic distributions are close prior art.

Defensible positioning. ALIEN-DL is an exploratory implementation family that tests whether a resource-allocation state can be coupled to (a) short-term transmission recovery, (b) efficacy adaptation and (c) structural persistence in a single synapse-centric classifier. The distinct hypothesis is the joint, implementation-level coupling and its factorial evaluation - not the invention of resource competition, plastic synapses, dendritic compartments, structural plasticity or exponentiated-gradient learning. The current tests do not yet validate the predicted interaction.

16 · REPRODUCIBILITY AND ARTIFACTS

Repository map, protocols and rerun notes

All source references below point to the public research-trials folder inspected for this report.

Artifact	Path / purpose
Research-trials root README	Documents both trials, summarized results and known caveats.
Trial A source	`A_xylia_prototype/alien_dl.py` - model; `run_benchmark.py` - training and benchmark runner.
Trial A data	`benchmark_results.csv`, `benchmark_summary.csv`, `factorial_results.csv`. CSV is authoritative for per-seed Trial A values.
Trial A diagnostics	`test_alien_dl.py`, `EXPERIMENT_REPORT.md`, `README__17_.md`.
Trial B source	`B_lite_trial/code/alien.py`, `run_alien.py`, `eta_sweep.py`, `unit_tests.py`.
Trial B result bundle	`ALIEN_DL_Research_Trials.zip` contains JSON results/logs according to the root README; extracted per-seed Trial B JSONs are not separately surfaced in the folder.
Dependencies	Root `requirements.txt` gives version lower bounds, not a full lockfile.

Trial A command pattern. From `A_xylia_prototype/`: `pip install -r requirements.txt`; `python run_benchmark.py`; `python -m unittest -v test_alien_dl.py`. The script defaults to datasets Wine, Breast Cancer and Digits; seeds 7, 21, 42; 100 epochs; Wine factorial seeds 7, 21, 42.

Trial B command pattern. From `B_lite_trial/code/`, run `python run_alien.py digits 40 3` or `python run_alien.py breast_cancer 40 3`. The root README notes `eta_sweep.py` selected on test data; that sweep is exploratory. `unit_tests.py` tests simplex normalization, positivity and a weighted-sum limiting case.

Reproducibility requirements for a v2 run. Pin exact dependencies; record OS, CPU/GPU, Python and library versions; export every seed-level prediction and metric; write split indices; save validation-selected hyperparameters; and store model configurations and checkpoints. Include a checksum manifest for the ZIP and extracted files so they cannot silently diverge.

Artifact integrity note

The repository contains both extracted files and ZIP bundles. The extracted source + Trial A CSVs are the readable authoritative record used here. The report treats Trial B summary values as repository-reported, because the bundled per-seed JSON/logs were not separately available as individual extracted files in the inspected folder.

17 · APPENDIX A

Formal comparison protocol for the next study

A stronger experiment should predefine common mechanics and align the two code paths before interpreting interaction effects.

Common allocation definition. For each domain k, define a positive mass $b_{e,k}$ and a fixed budget B_k. A structural event must redistribute mass while preserving the same total:

Shared fixed-ledger hypothesis

$$b_{e,k} \geq 0;\ \Sigma_{e \in S_k}\, b_{e,k} = B_k;\ \pi_{e,k} = n_k b_{e,k} / B_k$$

Use a constrained update with a projection or normalized exponentiated-gradient step. The chosen optimizer must state whether it is (i) a direct gradient in b, (ii) a gradient in logits, or (iii) a continuous replicator flow. Those are related, but not interchangeable implementation descriptions.

Matched coupling factors. R must only control allocation dependence of recovery; P must only control allocation dependence of local efficacy updates; A must only control allocation dependence of structural survival. The allocation state itself must be trained under the same rule across all eight conditions so that “coupling off” does not inadvertently turn off a different factor.

Primary contrast. For accuracy A_{rpa} and the three binary factors, define the average three-way interaction contrast over all matched cells as:

Factorial three-way contrast

$$I_{RPA} = A_{111} - A_{110} - A_{101} - A_{011} + A_{100} + A_{010} + A_{001} - A_{000}$$

Report the seed-wise contrast distribution and confidence interval, not only a single point estimate. Use the same test examples across all eight cells within each seed, and use multiple independent seeds. For inference, model dataset/task and seed variation; predefine correction for multiple contrasts.

Secondary measures. Report accuracy and macro-F1, calibration, wall-clock time, peak memory, FLOPs, active synapses, allocation Gini, entropy/effective fan-in, topology turnover, state-retention time and number of trainable parameters. A “win” based only on test accuracy is insufficient.

REFERENCES

References · biological and mathematical foundations

Bibliographic details and links checked during preparation on 14 November 2026.

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[5] Tsodyks, M. V., & Markram, H. (1997/1998 model lineage). Short-term synaptic dynamics are represented in the prototypes through resource-depression and facilitation variables; the implementations are simplified deterministic analogues, not full event-driven Tsodyks-Markram simulators. See also Markram et al. (1998) [2].

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Reference-use note

Biological references support the motivation and the existence of related mechanisms. They do not validate the computational proxies used by ALIEN-DL, and they are not evidence that the prototypes are biologically faithful.

REFERENCES

References · prior ANN architectures and datasets

Recent adjacent work is included to bound the novelty claim and guide the next comparison.

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[13] scikit-learn developers. `load_wine`, `load_breast_cancer`, and `load_digits` dataset documentation. Wine; Breast Cancer; Digits. Dataset sizes and feature counts cited in the report follow these pages.

RESEARCH STATUS

ALIEN-DL v0.1 is a reproducible starting point, not a solved architecture. The next decisive experiment is not a larger static classifier; it is a persistent-state temporal benchmark with matched controls, fixed budget accounting and preregistered evaluation.

Repository: The-Invention-Archive / ALIEN_DL_Research_Trials_2026.11.14